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graphia

Between Graph Semantics and Musicological Intent

Interpretive mediation in researcher, knowledge graph and LLM interaction

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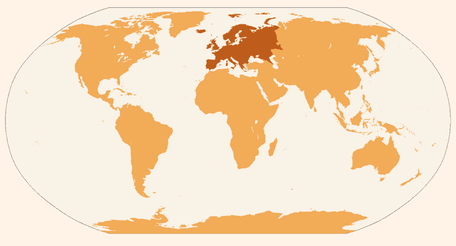
LLMs do not simply translate between musicologists and knowledge graphs. They mediate interpretive choices in both directions. Those choices need to remain inspectable.

Example: LinkedMusic. Are a cappella dance songs rare?

A denominator changed the answer

The Lomax dataset contains songs but no purely instrumental tunes. In many cultures, people dance to instrumental music, so the available dance-song denominator overrepresents vocal music.

Share of recorded dance songs that are a cappella



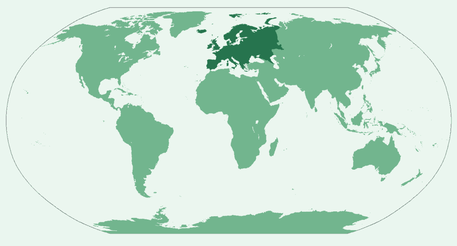
28.0% worldwide

48.7% Europe

Song-only data inflate this percentage.

INITIAL SCOPE

Share of cultures with ≥1 a cappella group dance song



6.7% worldwide · 58/870

4.7% Europe · 7/148

Only 58 of 870 cultures have such a recording.

REVIEWED REPAIR

Graphia Agent · LinkedMusic

RESEARCHER
Can you now search for all dance related songs in GJB and tell me the proportions - how common are a cappella dance songs without instruments?

AGENT
Total dance songs: 853
A cappella dance songs: 240
Proportion a cappella: 28.1%

RESEARCHER
Can you count the cultures in Europe which have recordings of a cappella dance group songs, and give me the percentage of these among all European cultures in the dataset

AGENT
Total distinct cultures with any songs in European countires : 148
Distinct cultures with ≥1 a cappela group dance song: 7 (4.7%)

Two directions

The same mediation problem runs in both directions

The researcher supplies the information need and judges the result. The LLM translates between language and graph operations. The graph constrains what the system can ask and what its answer can support.

1 MUSICOLOGICAL INTENT → 2 LINGUISTIC FORMULATION → 3 GRAPH MODEL + QUERY → 4 EVIDENCE + CLAIM

Researcher judges and repairs
LLM mediates transitions

GRAPHIA AGENT TESTING EXPERIMENT

Musicologists posed real questions. The agent selected a knowledge graph, generated SPARQL and returned claims for expert review.

Russian opera: a proxy becomes a false claim

QUESTION	Find Russian operas in the CKG.
BOUNDARY	The graph lacks a country relation, so known titles become a search proxy.
BREAKDOWN	“Romeo and Juliet by Tchaikovsky” conflates his overture with Prokofiev’s ballet.
REPAIR	Expert review corrects the attribution and limits the claim.

Kotzebue: the graph exposes a mystery

OBSERVATION	One person is represented more than anyone else, across records extending far beyond his lifetime.
QUESTION	Who is he—and why is he so prominent? We had never heard of him.
BOUNDARY	The CKG reveals the anomaly but cannot explain the person–work roles that produced it.
RESOLUTION	August von Kotzebue was an opera librettist who dominated the German-language stage of the period. LinkedMusic/RISM restores role-specific evidence of his musical reception.

MUSPARQL Musparql records multiple reviewed formulations for each SPARQL query. Ambiguity and pragmatic intent remain visible rather than being collapsed into one ideal wording.

Release or album?

CLOSER TO QUERY	Which releases include a performance by X?
MORE PRAGMATIC	Which albums include a performance by X?
TENSION	The pragmatic wording changes the entity: an album can have more than one release.

Same fingerprints or duplicate tracks?

CLOSER TO QUERY	Which tracks have the same fingerprints?
PRAGMATIC INTENT	Which tracks are duplicates?
TENSION	“Duplicates” requires knowledge of how the graph operationalises duplication through fingerprints.

MUSPARQL: HOW IT OPERATIONALISES THE TENSION

BENCHMARK CONSTRUCTION

Review establishes faithfulness to the information need. Musparql records alternative formulations with full provenance: source records, model outputs, reviewer decisions, canonical selections, alternative formulations and repair history. Ambiguity and pragmatic intent remain inspectable.

LINGUISTIC DIMENSIONS EXPERIMENT

The study uses alternative formulations and Musparql’s ambiguity model to explore where naturalness, pragmatism and room for interpretation diverge. ***The interface is implemented. The main study has not yet run.***

FROM CASES TO EVALUATION

Why this leads to a benchmark

A single question, query or answer conceals the transformations that produced it. Endpoint accuracy alone cannot distinguish faithful translation from a plausible workaround, an unsupported claim or a successful repair.

A mediation-aware benchmark preserves

Query and formulations
Provenance and reviewer decisions
Alternatives and repair history

Its practical purpose

The benchmark tests whether a system preserves the researcher’s information need and exposes graph boundaries. It also records how the system qualifies evidence and whether a reviewer can repair the result.

Making interpretation visible allows researchers to challenge and revise it.

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HERMES CASE STUDIES
odoma-ch.github.io/hermes-musigraphia-report/explorations/

MUSPARQL
github.com/ppquadrat/musparql-aligner